

# INDIAN STATISTICAL INSTITUTE

Project Linked Person — Selection 2026–2027

Practical Examination

Total: 14 + 16 + 10 = 40 marks      Duration: 90 minutes

## EXAMINATION INSTRUCTIONS

1. You have root on a virtual machine. It is connected to no real network. Log in with `ssh candidate@192.168.56.10`.
2. Each question has four parts: *the problem*, *how to reproduce it*, *what to fix*, and *how to check your work* (so that you do not have to guess whether you are finished).
3. Partial credit would be given if you can explain the problem and the reason behind, even if you cannot fix it.

### Q1. Disk full on the lab file server.

[14 marks]

*Reported by Debarshi, research scholar:* “I cannot copy anything into `/srv/data`, it says the disk is full, and nothing I can see in there accounts for that much space. Submitting a job into `/srv/spool` *also* says the disk is full, but `df -h` shows that one has plenty of space left. That part makes no sense to me.”

**Background:** this server has three separate disks. `/srv/data` and `/srv/spool` look like ordinary folders, but each one is really its own small disk, attached at that path (a *mount point*). The rest of the system lives on a third, much larger disk.

|                         |                         |             |
|-------------------------|-------------------------|-------------|
| <code>/</code>          | the system disk         | about 24 GB |
| <code>/srv/data</code>  | the project data disk   | 1 GB        |
| <code>/srv/spool</code> | the job-submission disk | 48 MB       |

Each disk fills up on its own: one being full says nothing about the others. `df -h / /srv/data /srv/spool` prints one line per disk.

Though the two complaints look alike, they are **two unrelated faults on two different disks**, with different causes. Fixing one does nothing for the other.

#### (a) `/srv/data` is full

[11 marks]

`/srv/data` is at 100%, and the files you can see there do not add up to the space reported as used.

*Reproduce:*

|                                    |                                       |
|------------------------------------|---------------------------------------|
| <code>df -h /srv/data</code>       | # 100% used                           |
| <code>sudo du -sh /srv/data</code> | # much smaller than df says is in use |

*Fix:* bring `/srv/data` back to **under 10% used** (the `Use%` column of `df -h`). There is more than one reason this disk is full: freeing the largest obvious files will not finish the job. Consider the differences shown in the output of `df` and `du`. Explain the reason for the difference.

*Check:*

```
df -h /srv/data          # Use% under 10%
ls /srv/data/current     # five survey-block CSV files, still there
```

**(b) /srv/spool refuses new files [3 marks]**

New files cannot be created on `/srv/spool`, *although `df -h` reports free space on it*. Something other than free space has run out. This has nothing to do with `/srv/data`.

*Reproduce:*

```
df -h /srv/spool          # shows space free
touch /srv/spool/incoming/testjob  # "No space left on device"
```

*Fix:* make `/srv/spool` able to accept new files again. It is *not* short of space: freeing bytes there will not help. A filesystem can run out of more than one kind of resource, and `df` has another mode that reports the other one. Explain why `/srv/spool` reported “No space left on device” while `df -h` showed free space.

*Check:*

```
touch /srv/spool/incoming/t && echo OK && rm /srv/spool/incoming/t
```

**Constraints**

- Nothing under `/srv/data/current` may be deleted or changed. It is live survey data with no second copy on this machine.
- `/srv/data/archive/README.txt` says what is safe to remove there. Read it before deleting anything.

**Q2. Shared project folder unusable. [16 marks]**

*Reported for the ACMU Lab group:* “Buddhadev cannot save anything into `shared/`, and cannot open `notes.md` at all. When I create a file there, the others cannot edit it — it comes out belonging to my own refused. And Chandrima from the internal audit cell needs to *read* that folder for the annual review. He cannot see anything.”

**Background: who is who.** `/srv/projects/acmu/` (and `shared/` inside it) belongs to the `acmu` group. Anyone who is not a member or the auditor should have no access at all.

|                        |                              |                                |
|------------------------|------------------------------|--------------------------------|
| <code>arnab</code>     | member of <code>acmu</code>  | read and write everything      |
| <code>buddhadev</code> | member of <code>acmu</code>  | read and write everything      |
| <code>chandrima</code> | auditor, <b>not</b> a member | read everything, write nothing |

To see the current state:

```
cd /srv/projects/acmu
stat -c '%a %U:%G %n' . shared shared/notes.md
id buddhadev; getfacl /srv/projects/acmu/shared
```

The complaints overlap, but underneath them are **five independent faults**, each with its own cause and fix. Fixing one does nothing for the others; each is marked on its own. Tackle them in any order. Test as the users named, not as root — root can do everything, so a check run as root proves nothing.

**(a) Buddhadev is no longer in the acmu group** [4 marks]

*Reproduce:* `id buddhadev` — `acmu` is missing. *Fix:* put him back, without changing any of his other groups.

```
id -nG buddhadev | grep -qw acmu && echo "buddhadev is a member"
```

**(b) The shared folder does not let the group write** [7 marks]

Members of `acmu` cannot create files in the folder at all. This is about the permission bits on the two directories, `/srv/projects/acmu` and `shared/`. Arnab is still a member, so test as her.

```
sudo -u arnab touch /srv/projects/acmu/shared/t1      # Permission denied
stat -c '%a %U:%G %n' /srv/projects/acmu /srv/projects/acmu/shared
```

*Fix:* both directories must let the `acmu` group read, write and enter them, and give everyone else no write access. Remember that a directory needs its execute bit before anyone can go into it at all.

```
sudo -u arnab touch /srv/projects/acmu/shared/t1 && echo "write OK"
sudo rm -f /srv/projects/acmu/shared/t1
```

**(c) notes.md has become private** [3 marks]

`shared/notes.md` is the group's working file, but now belongs to one person and only she can read it. *Fix:* group `acmu`, readable and writable by the group, no access for anyone else.

```
stat -c '%a %G' /srv/projects/acmu/shared/notes.md    # 660 acmu
```

**(d) The auditor cannot read the folder** [2 marks]

Chandrima's access came from per-user entries on the files themselves, and those have been wiped. *Fix:* give him read-only access to everything under `/srv/projects/acmu/`, including files created there in future — *without* adding him to the group.

```
sudo -u chandrima cat /srv/projects/acmu/shared/notes.md # must work
sudo -u chandrima touch /srv/projects/acmu/shared/t2      # must FAIL
```

**When all five are done**

Buddhadev's original complaint needs (a), (b) and (c) together:

```
sudo -u buddhadev touch /srv/projects/acmu/shared/t3      # must work
sudo -u buddhadev cat /srv/projects/acmu/shared/notes.md  # must work
```

**Constraints — these apply to every part, and are marked**

- **Chandrima must not be put into the acmu group.** Audit policy forbids putting auditors into the groups they audit. He must be able to read and must *not* be able to write.
- No `chmod 777`, anywhere. This is a multi-user machine.
- Do not change anyone's login shell, password or home directory.

*Hint.* Group membership changes do not affect a shell that is already open; `sudo -u user` starts a fresh one.

**Q3. Cannot log in to the ACMU lab server.**

[10 marks]

*Reported by Debarshi, research scholar:* “I am supposed to log in to the lab server as `acmusrv`. It worked last week. Now it just says *Permission denied*, and then asks me for a password I was never given. I have not touched anything since.”

`ssh acmulab` should log you in to the account `acmusrv` on the lab server using the key `~/.ssh/id_acmu`, **without a password**. Two things are wrong, and they produce the **same message**: `Permission denied (publickey)`. Identify both of them.

**How to reproduce it**

```
ssh acmulab

ssh -v acmulab 2>&1 | grep -i 'identity file\|offering\|publickey'
cat ~/.ssh/config
ls -l ~/.ssh/
```

**What to fix**

[10 marks]

**3.1 Make the acmulab alias use the right key.** The `~/.ssh/config` file is corrupted.

Fix it so that `ssh acmulab` succeeds without a password prompt.

**3.2 Make the key usable.** Find out why `ssh` still refuses to use the key, and fix it.

**How to check you have fixed it**

```
ssh -o BatchMode=yes acmulab whoami      # prints: acmusrv
ls -l ~/.ssh/id_acmu                     # -rw----- (0600)
```

`BatchMode=yes` forbids any password prompt, so if this prints `acmusrv` the key really is doing the work.